

Hand calculations for Problem 1.

Input shape = (B, channels, height, width)

where B is batch size

$$= (B, 3, 32, 32)$$

$$\text{output size} = \left\lceil \frac{N - F + 2P}{S} \right\rceil + 1$$

Conv1:  $F=3, S=1, P=1$ , output channels = 32

$$\text{output size} = \left\lceil \frac{32 - 3 + 2}{1} \right\rceil + 1 = 32$$

$$\text{Shape} = (B, 32, 32, 32)$$

Pool 1:  $F=2, S=2, P=0$

$$\text{output size} = \left\lceil \frac{32 - 2 + 0}{2} \right\rceil + 1 = 16 \Rightarrow \text{Shape} = (B, 32, 16, 16)$$

Conv 2:  $F=3, S=1, P=1$ , output channels = 64

$$\text{output size} = \left\lceil \frac{16 - 3 + 2}{1} \right\rceil + 1 = 16 \Rightarrow \text{Shape} = (B, 64, 16, 16)$$

Pool 2:  $F=2, S=2, P=0$

$$\text{output size} = \left\lceil \frac{16 - 2 + 0}{2} \right\rceil + 1 = 8 \Rightarrow \text{Shape} = (B, 64, 8, 8)$$

Conv 3:  $F=3, S=1, P=1$ , output channels = 128

$$\text{output size} = \left\lceil \frac{8 - 3 + 2}{1} \right\rceil + 1 = 8 \Rightarrow \text{Shape} = (B, 128, 8, 8)$$

Pool 3:  $F=2, S=2, P=0$

$$\text{output size} = \left\lceil \frac{8 - 2 + 0}{2} \right\rceil + 1 = 4 \Rightarrow \text{Shape} = (B, 128, 4, 4)$$

Flatten:

channels  $\times$  height  $\times$  width

$$128 \times 4 \times 4 = 2048 \Rightarrow \text{Shape} = (B, 2048)$$



RF calculations for Problem 1

$$RF_{out} = RF_{in} + (k-1)j_{in} \quad j_{out} = j_{in} \times s$$

input:  $RF_{out} = 1, j_{out} = 1$

conv1:  $k=3, s=1, j_{in}=1, RF_{in}=1$

$$RF_{out} = 1 + (3-1) \times 1 = \boxed{3} \quad j_{out} = 1 \times 1 = \boxed{1}$$

pool1:  $k=2, s=2, j_{in}=1, RF_{in}=3$

$$RF_{out} = 3 + (2-1) \times 1 = \boxed{4} \quad j_{out} = 1 \times 2 = \boxed{2}$$

conv2:  $k=3, s=1, j_{in}=2, RF_{in}=4$

$$RF_{out} = 4 + (3-1) \times 2 = \boxed{8} \quad j_{out} = 2 \times 1 = \boxed{2}$$

pool2:  $k=2, s=2, j_{in}=2, RF_{in}=8$

$$RF_{out} = 8 + (2-1) \times 2 = \boxed{10} \quad j_{out} = 2 \times 2 = \boxed{4}$$

conv3:  $k=3, s=1, j_{in}=4, RF_{in}=10$

$$RF_{out} = 10 + (3-1) \times 4 = \boxed{18} \quad j_{out} = 4 \times 1 = \boxed{4}$$